# PFGM-IQA: CT Image Quality Assessment using Poisson Flow Generative Models

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*Abstract*—**Image Quality Assessment (IQA) is essential for optimizing radiation dose in Computed Tomography (CT) while ensuring radiologists achieve the highest diagnostic accuracy. To advance research in this field, we propose a novel algorithm for CT image quality assessment without the need for reference images. To overcome the challenge of no-reference IQA in CT scans, we employ a novel approach using Poisson Flow Generative models (PFGM) to generate pseudo-reference images for low-dose CT scans. These pseudo-reference images, paired with corresponding low-dose inputs, are fed into a robust regression network specifically designed for this task. To enhance feature extraction, we design a nested convolutional architecture using multi-scale features to improve the feature extraction capability of the regression Network. Furthermore, to strengthen the monotonic correlation between subjective and objective scores, we incorporate the relative distance information within each batch and enforce the relative ranking among the images. To propel research in this domain, we also build and release a new abdominal CT dataset with labeled IQA scores, tailored for benchmarking IQA methods. Extensive qualitative and quantitative experiments, conducted on both public datasets and our newly released dataset, demonstrate that our proposed PFGM-IQA method significantly outperforms state-of-the-art techniques in CT Image Quality Assessment.**

*Index Terms*—**CT Image quality assessment, Poisson Flow Generative model, Nested Convolutional Network**

## I. Introduction

Computed tomography (CT) is one of the most commonly used medical imaging techniques and plays an integral role in the diagnostic process for physicians. However, the high dose of radiation exposure during CT scanning can cause some damage to the patient's body, which has a greater health risk. Therefore, imaging with lower radiation doses has become an important technique to mitigate the risk of radiation exposure. However, the radiation reduction of low-dose CT reduces image quality, often introducing increased noise and artifacts. These degradations not only compromise a physician's ability to make accurate diagnoses but also hinder downstream image-processing tasks, such as image segmentation [1], restoration [2], and enhancement [3]. Therefore, advancing precise Image Quality Assessment (IQA) methods is essential for optimizing radiation doses in CT, ensuring both patient safety and the reliability of diagnostic outcomes.

For a given image, the goal of IQA is to predict a score that is as close as possible to the human mean subjective score using an image quality assessment algorithm. IQA are typically classified into full reference image quality assessment (FR-IQA), reduced reference image quality assessment (RR-IQA), and no reference image quality assessment (NR-IQA) depending on whether a reference image is required or not. The FR-IQA methods derive quality scores by comparing both the reference and distorted images, while RR-IQA methods use distorted image information coupled with partial reference image information. Due to their reliance on a reference image for calculating image quality scores, FR- and RR-IQA methods have demonstrated promising performance. Consequently, the peak signal-to-noise ratio (PSNR) and structural similarity index measure (SSIM) are widely adopted as standard evaluation measures for image quality owing to their computational efficiency and strong correlation with human visual perception in traditional distortion types, such as blur, compression, and Gaussian noise. However, in practical applications-especially in the medical field-reference images are often unavailable, limiting the effectiveness of FR- and RR-IQA methods. Obtaining high-quality reference images in medical contexts can pose risks to patients, making it impractical. As a result, NR-IQA methods, which do not require reference images, are critically important for advancing medical imaging, offering a safer and more viable solution for accurate image quality assessment in healthcare settings.

Traditional NR-IQA methods mainly rely on manually crafted features designed to simulate the perceptual characteristics of the human visual system (HVS). After preprocessing the image, various manual features are extracted from the image, including statistical features (mean and variance), texture features (Gabor filter), edge features (Sobel operator), structural features (local binary pattern), and frequency domain features (Fourier transform). These features are then fused using various techniques to extract the statistical distribution of the features. Finally, a mapping function, such as support vector regression (SVR) [4], is designed to predict the quality of the input image. However, due to the complexity of image content and distortion patterns, the representation ability of manual features is often unsatisfactory.

Recently, convolutional neural networks (CNNs) have garnered significant attention in the realm of no-reference image quality assessment (NR-IQA) tasks due to their powerful capabilities, such as the weighted average deep image quality metric (WaDIQaM) [5], multi-task end-to-end optimized deep neural network (MEON) [6], image quality transformer (TRIQ) [7], and multi-dimensional attention network for NR-IQA (MANIQA) [8]. However, the lack of reference information poses a challenge that makes these methods always have bottlenecks compared to FR-IQA and may adversely affect their overall performance.

Therefore, some recent methods adopt reconstruction-based strategies to learn quality-aware features from raw images. Notably, frameworks leveraging generative adversarial networks (GANs) [9] have emerged. For example, a quality-aware GAN is developed to generate hallucinated references conditioned on distorted images [10]. Similarly, other studies propose innovative solutions, like generating the main content of distorted images [11] and developing restorative adversarial networks to reconstruct input images [12]. Recently, D-BIQA [13] introduces a conditional denoising diffusion model to generate the main content of low-quality images. However, while these generative models excel at addressing quality degradations in natural images, they often struggle when applied to CT images due to the distinct characteristics of CT-specific noise. Unlike natural image noise, which is typically additive or Gaussian, noise in CT images arises from quantum noise, governed by a Poisson distribution. This quantum noise, caused by random photon fluctuations, introduces granular artifacts that significantly challenge traditional generative approaches.

To overcome these challenges, we propose a novel CT image quality assessment based on Poisson flow generative models, named PFGM-IQA. PFGM-IQA integrates two innovative components: (1) **Poisson flow generative models (PFGM)** to generate high-fidelity pseudo-reference images tailored to the Poisson noise present in CT scans, and (2) a powerful **hierarchical feature fusion regression network** that combines the strengths of CNNs for capturing local details and transformers for modeling global contextual relationships. There are two main factors that affect the quality of CT images, noise and artifacts, among which quantum noise follows a Poisson distribution. To this end, we use Poisson flow generative models to generate corresponding pseudo-reference images to make up for the lack of reference images and help the subsequent score regression model capture detailed features. The second component is the Hierarchical feature fusion regression network. We adopt a ladder structure to fuse the upper-level features with the lower-level features so that the model can make full use of features of different scales. At the same time, the features extracted by the last layer of CNN are passed into the Vision Transformer (ViT) in turn, and the obtained global features are fused with the local features to predict the final image quality. We conduct comprehensive experiments on both the publicly available datasets and our newly synthesized dataset. Extensive experimental results demonstrate that the proposed method achieves superior performance compared to existing IQA methods. The main contributions of this paper are summarized as follows:

- We introduce the Poisson Flow Generative Models to create high-quality pseudo-reference images, which can significantly enhance the subsequent score regression model to capture intricate features.
- We propose a novel hierarchical feature fusion regression network, which seamlessly integrates ViT and CNN modules to fuse non-local features with local features to deliver highly accurate predictions of final image quality.
- We leverage the proposed Poisson Flow Generative Models to synthesize a comprehensive dataset tailored for CT quality assessment, which can serve as a new resource for advancing research in this field.

## II. Related Work

### A. Natural Image Quality Assessment

As deep neural networks have shown strong learning capabilities in recent years, the using of deep CNNs to blindly evaluate image quality has become a trend. Yan *et al.* [14] propose a two-stream convolutional network, which includes two subcomponents, image and gradient image, to capture different levels of input information. Su *et al.* [15] develop an adaptive super network to aggregate local distortion features and global semantic features. Zhu *et al.* [16] propose a meta-learning framework for IQA to improve generalization to unseen image distortions. Ke *et al.* [17] propose a multi-scale image quality transformer (MUSIQ), which uses a transformer architecture to solve the problem of images with different sizes and aspect ratios. Pan *et al.* [18] propose an NR-IQA method based on the visual compensation restoration network, which uses a non-adversarial model to effectively handle the restoration task of distorted images and accurately establish the quality reconstruction relationship between the distorted image and its restored image. There are also some methods [8] that incorporate attention mechanisms into CNNs, significantly enriching the representation of local fine-grained details while capturing multi-scale global semantic information, further advancing the state-of-the-art in IQA methods.

### B. Medical Image Quality Assessment

Although the research and development of natural image IQA methods has been very rapid, these techniques face notable limitations when applied to the unique and complex domain of medical imaging. In addition, for natural image IQA, there are many publicly available datasets with human annotations, including LIVE [19], TID2013 [20], and PIPAL [21]. However, there is currently no equivalent dataset for medical IQA, and it is impossible to obtain sufficient data volume and annotations for training. This scarcity makes it challenging to train robust IQA models for medical applications and highlights the urgent need to develop novel methods specifically tailored for medical IQA. Due to the different characteristics of medical imaging, there are many different methods targeting specific imaging modalities. These include MRI [22], ultrasound imaging [23], CT [24], X-ray [25] and other medical image types.

Recently, many researchers have explored the field of CT image quality assessment. Pal *et al.* [24] proposes a no-reference IQA model via jointly learning noise level estimation and image quality quantification. Song *et al.* [26] introduce semi-supervised learning and used pseudo labels of unlabeled data to guide model training. Ahamed *et al.* [27] use natural images of similar tasks for transfer learning. Some methods use task-based IQA models [28], [29] to detect virtual inserted objects with simple geometric shapes and calculate the quantitative quality of CT images. Shi *et al.* [13] propose a conditional DDPM to predict the main content of distorted

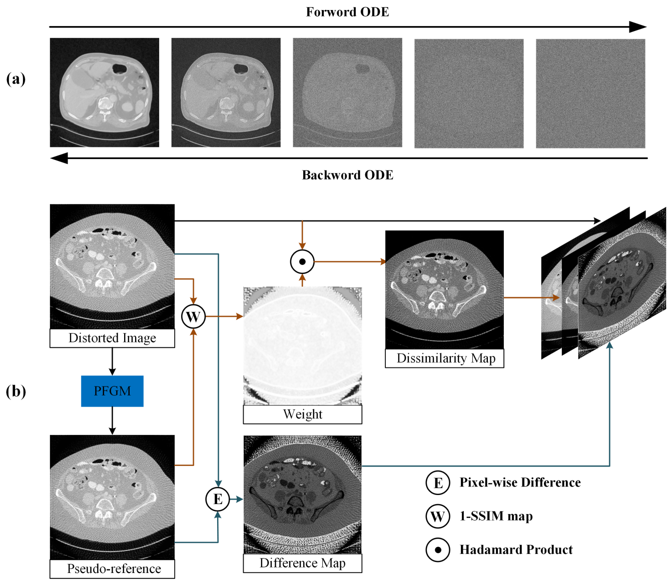


Fig. 1. (a) The workflow of the PFGM that maps low-dose CT images to normal dose counterparts. (b) The three-channel image is obtained by processing the distorted image and the pseudo-reference image generated by PFGM.

images, which are then processed and input into the quality evaluator to predict the image quality of low-dose CT.

## III. Method

### A. Poisson Flow Generative Models

Diffusion and Poisson Flow Models are emerging deep generative models that have demonstrated exceptional performance across a variety of tasks, successfully generating images with low noise levels while preserving critical details. Diffusion models [30] are originally inspired by nonequilibrium thermodynamics. They initially transform the data distribution into a noise distribution by iteratively adding Gaussian noise, then learn to reverse this process, gradually removing the noise to generate clear images. In contrast to fractional-based diffusion models, PFGM [31]–[33] focus on the high-dimensional electric field rather than estimating a time-dependent fractional function. PFGM generates a uniform distribution of negative charges over a hemisphere and subsequently tracks their motion back to the $z = 0$ plane, where they are distributed as the data distribution. Poisson flow can be viewed as a continuous normalized flow [34], as it continuously maps between arbitrary distributions and easily sampled distributions: an n-dimensional Gaussian distribution in previous work and a uniform distribution over the n-dimensional hemisphere in PFGM. In practice, PFGM implements Poisson flow by solving a pair of direct and inverse ordinary differential equations (ODEs) induced by the electric field given by an n-dimensional version of Coulomb's law (the gradient of the solution of the Poisson equation with the data as the source) [32].

The pseudo-reference image generated by the Poisson Flow Generative Model (PFGM) plays a pivotal role as a surrogate for real reference images when assessing low-quality CT scans. The smaller the gap between the pseudo-reference and the real reference, the more accurate the performance of the quality regression network. Given that Poisson noise in CT images is a discrete noise type whose variance scales with signal intensity, PFGM is particularly well-suited for modeling these noise characteristics. Unlike traditional approaches, PFGM excels in capturing the statistical behavior of Poisson noise with remarkable accuracy, effectively reducing artifacts often introduced by conventional denoising methods. This capability underscores its potential to revolutionize quality assessment in medical imaging.

Building on the foundational principles of PFGM, the single-step posterior sampling Poisson Flow Generative Model [33] introduces an innovative approach by hijacking and regularizing the sampling process. Instead of running all the way from a sample from the prior noise distribution, the single-step posterior sampling Poisson flow generative model will hijack the sampling process at $\tau, \tau < T$ by simply inserting the condition image. Therefore, we also adopt this approach to achieve excellent denoising performance without the need for intensive computational resources, significantly accelerating the image generation process. This enhanced efficiency not only optimizes resource utilization but also makes it an ideal solution for our CT image quality assessment task, striking a perfect balance between performance and practicality.

PFGM interprets the N-dimensional data $x \in \mathbb{R}^N$ as electric charges in an N+D-dimensional space augmented with an extra dimension z: $\tilde{x} = (x, z) \in \mathbb{R}^{N+D}$ and $D \in \mathbb{Z}^+$. In particular, the training data are placed on the $z = 0$ hyperplane, and the electric field lines emitted by the charges define a bijection between the data distribution and a uniform distribution on the infinite hemisphere of the augmented space. To perform generative modeling, PFGM learns the following high-dimensional electric field, which is the derivative of the electric potential in a Poisson equation:

$$E(\tilde{x}) = \frac{1}{S_{N+D-1}(1)} \int \frac{\|\tilde{x} - \tilde{y}\|}{\|\tilde{x} - \tilde{y}\|_{N+D}} p(y)\, dy, \qquad (1)$$

where $p(y)$ is the ground truth data distribution, $S_{N+D-1}(1)$ is the surface area of the unit $(N + D - 1)$-sphere, and $\tilde{y} = (y, 0) \in \mathbb{R}^{N+D}$ and $\tilde{x} = (x, z) \in \mathbb{R}^{N+D}$ the augmented ground truth and perturbed data, respectively.

Ordinary Differential Equation (ODE) $p(y)$ defines a surjection from a uniform distribution on an infinite $(N + D)$-dim hemisphere. and the data distribution on the N-dim $z = 0$ hyperplane. Moreover, for any positive $r$, the mapping has the symmetry on the D-dim cylinder $\sum_{i=1}^{D} z_i^2 = r^2$. Therefore, we only need to keep track of the norm of the augmented variable. Specifically, take note of $dz_i = E(\tilde{x})z_i$, and observe that the time derivative of $r$ is

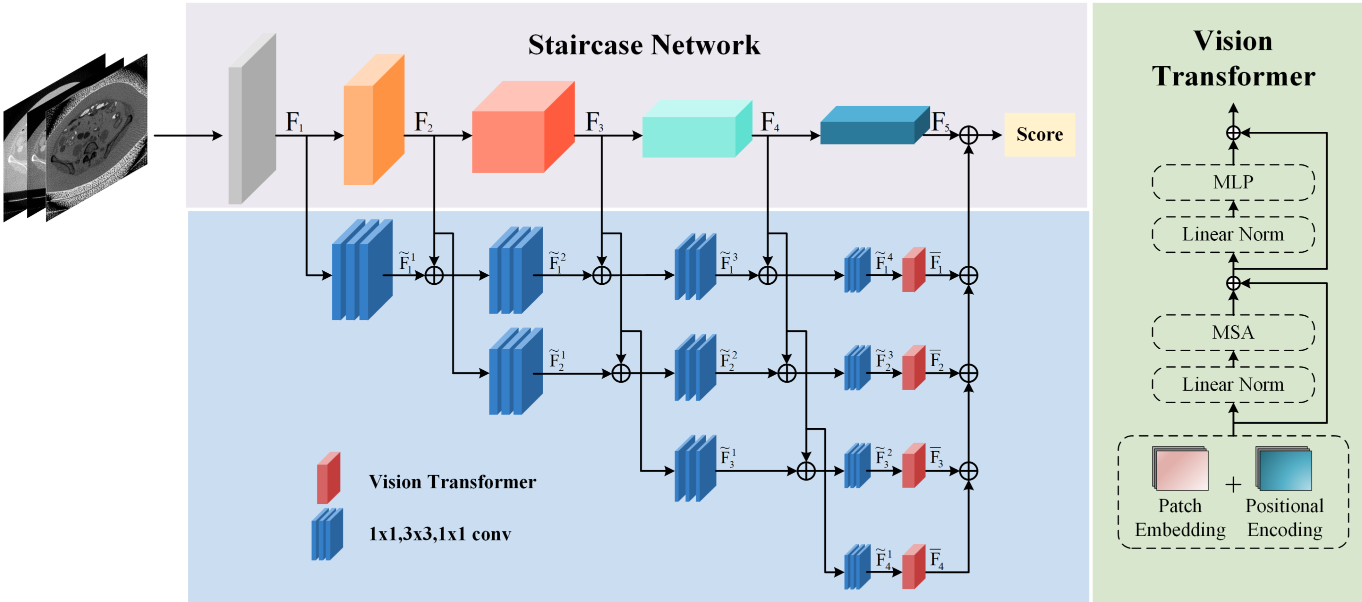


Fig. 2. The overall architecture of the proposed PFGM-IQA. Low-dose CT images are processed by PFGM to generate corresponding high-quality pseudo-reference images. After processing, the multi-channel images are fed into the hierarchical feature fusion regression network for feature extraction.

$$\begin{aligned}\frac{dr}{dt} &= \sum_{i=1}^{D} \frac{z_i}{r}\frac{dz_i}{dt} \\ &= \int \frac{\sum_{i=1}^{D} z_i^2}{S_{N+D-1}(1) r \|\tilde{x} - \tilde{y}\|_{N+D}} p(y)\, dy \\ &= \frac{1}{S_{N+D-1}(1)} \int \frac{r}{\|\tilde{x} - \tilde{y}\|_{N+D}} p(y)\, dy.\end{aligned} \tag{2}$$

Based on the above, we can use $r$ as the anchor variable in the ODE $dx/dr$ by change-of-variable:

$$\frac{dx}{dr} = \frac{dx}{dt}\frac{dt}{dr} = \frac{E(\tilde{x})_x}{E(\tilde{x})_r} \tag{3}$$

Thus, we transform the previously mentioned projection problem into a bi-projection between the prior distribution that is easily sampled on the $r = r_{max}$ hyper-cylinder and the data distribution on the hyperplane at $r = 0$ (i.e., $z = 0$). In addition, a perturbation-based target is added to ensure that the minimizer in the following loss objective matches the true ground electric field in Eq. 3:

$$E_{r\sim p(r)}\, E_{y\sim p(y)}\, E_{x\sim p_r(x|y)} \left[ \left\| f_\theta(\tilde{x}) - \frac{x-y}{r/\sqrt{D}} \right\|_2^2 \right], \tag{4}$$

where $p(r)$ is the training distribution over $r$. Starting from the prior distribution $p_{r_{max}}$, and after training the neural network using the objective function in Eq. 4, we can then generate samples i.e., $dx/dr = E(\tilde{x})_x/E(\tilde{x})_r = f_\theta(\tilde{x})/\sqrt{D}$ using the ODE anchored at $r$ as described in Eq. 3.

The conditional image $c$ is fed as an additional input into the network to estimate the time-dependent score function.

**Algorithm 1** PFGM training

1: Sample a batch of data $\{y_i, c_i\}_{i=1}^{B}$ from $p(y, c)$
2: Sample standard deviations $\{\sigma_i\}_{i=1}^{B}$ from $p(\sigma)$
3: Sample $r$ from $p_r$ : $\{r_i = \sigma_i\sqrt{D}\}_{i=1}^{B}$
4: Sample radii $\{R_i = p_{r_i}(R)\}_{i=1}^{B}$
5: Sample uniform angles
$\{v_i = \frac{u_i}{\|u_i\|_2}\}_{i=1}^{B}$, $u_i \sim N(0, I)$
6: Get perturbed data $\{\hat{y}_i = y_i + R_i v_i\}_{i=1}^{B}$
7: Calculate loss
$\ell(\theta) = \frac{1}{B}\sum_{i=1}^{B} \lambda(\sigma_i)\|f_\theta(\hat{y}_i, \sigma_i, c_i) - y_i\|_2^2$
8: Update network parameters $\theta$ using Adam

This has been empirically proven to be very successful. PFGM also follows this strategy to transform from an unconditional generator to a conditional generator. The training process of PFGM is presented in Algorithm 1.

While Algorithm 1 provides a sufficient framework for obtaining the conditional generator, we further optimize the process to significantly enhance the sampling speed. Sepecifically, we start the sampling process at $e_i = \tau \in \mathbb{Z}^+, \quad \tau < T$ by directly inserting the conditional image $x_\tau = c$, instead of the original sampling from the prior noise distribution. In other words, we will mix $x_{i+1}$ with $x_\tau = c$, the input image we seek to denoise, using weight $w \in [0, 1]$. The sampling is shown in Algorithm 2. The specific process of PFGM is shown in Figure 1.

### *B. Hierarchical Feature Fusion Regression Network*

The images generated by PFGM can provide pseudo-references for the quality regression network to compensate

**Algorithm 2** PFGM sampling

Get initial data $x_\tau = c$
**for** $i = \tau, \ldots, T-1$ **do**
  $d_i = (x_i - f_\theta(x_i, t_i, c))/t_i$
  $x_{i+1} = x_i + (t_{i+1} - t_i) d_i$
  **if** $t_{i+1} > 0$ **then**
    $d_i' = (x_{i+1} - f_\theta(x_{i+1}, t_{i+1}, c))/t_{i+1}$
    $x_{i+1} = x_i + (t_{i+1} - t_i) \left(\frac{1}{2} d_i + \frac{1}{2} d_i'\right)$
  **end if**
  $x_{i+1} = w x_{i+1} + (1-w) x_\tau$
**end for**
**return** $x_T$

for the lack of real reference information. To fully harness the potential of these pseudo-references, we introduced the concepts of difference maps and dissimilarity maps, which enable more precise comparisons between the generated pseudo-references and low-dose CT images. These maps provide critical insights into image quality by quantifying variations and inconsistencies. The calculation formulas are as follows:

$$I_{\text{e}} = \left| I_d - G_\theta(I_d) \right|, \tag{5}$$

$$I_{\text{dis}} = I_{\text{e}} \left(1 - SSIM(I_d, G_\theta(I_d))\right), \tag{6}$$

where $G_\theta$ represents the processing of PFGM and $\theta$ represents the network parameters. SSIM represents the structural similarity between two images. We integrate the original low-dose CT image $I_d$, the discrepancy image $I_e$, and the dissimilarity maps $I_{dis}$ to form a comprehensive three-channel image $\mathbf{E} \in \mathbb{R}^{H \times W \times 3}$. Here, $H$ and $W$ correspond to the height and width of the image, respectively. This multi-dimensional representation is then used as input for the quality regression network, enabling it to leverage richer information for more accurate image quality assessment.

The widely used CNN models excel at progressively learning features from low-level details to high-level abstractions as network depth increases. However, in the human visual system (HVS), both low-level visual cues and high-level semantic information are indispensable for accurately perceiving image quality. To mirror this capability, we implement a ladder-like structure that hierarchically integrates features extracted from intermediate layers, enabling the model to fully utilize the rich spectrum of visual information spanning from fine-grained details to semantic context. Additionally, CT images often exhibit radial artifacts, a challenge that traditional CNNs struggle to address due to their inherent focus on local spatial structures. To address this limitation and improve the model's ability to understand non-local feature relationships, we incorporate a Vision Transformer after the final CNN layer. This integration effectively combines global contextual features with localized information, significantly improving the model's holistic understanding of image quality. The overall architecture of this quality assessment model, integrating hierarchical feature fusion and a Transformer layer, is illustrated in Figure 2.

The CNN architecture can be divided into $N$ stages according to the dimension of the feature map. $F_i$ represents the feature map extracted in the i-th stage, where $i \in [1, 2, \ldots, N]$. We aim to integrate the features extracted in each stage into the final feature representation, but the number of channels and dimensions in each stage are different. To solve this problem, we introduce a bottleneck structure consisting of three convolution operations [35]. The number of channels and resolution of the feature map $F_i$ are adjusted to an appropriate level through $1 \times 1$, $3 \times 3$, and $1 \times 1$ convolution layers in turn. The feature map $\tilde{F}_i$ is expressed as follows:

$$\tilde{F}_i = W_{1\times 1} W_{3\times 3} W_{1\times 1} F_i = W F, \tag{7}$$

where $W_{1\times 1}$ and $W_{3\times 3}$ represent the weight matrices of the $1 \times 1$ convolutional layer and the $3 \times 3$ convolutional layer respectively. To further improve the model's feature modeling capabilities, we introduced Transformer modules into the model. These modules include components such as position encoding, attention mechanism, and multi-layer perceptron (MLP). Among them, the attention mechanism is calculated as follows:

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right) V, \tag{8}$$

where query (Q), key (K), and value (V) projections are achieved by 3 independent linear projections. And $d_k$ indicates the spatial dimension of Q, K and V. We use the features $\tilde{F}_i$ from the last layer of the trapezoidal structure CNN as the input to the ViT, resulting in the output $\overline{F}_i$. Then, we integrate features from different layers:

$$F = \sum_{i=1}^{N-1} \overline{F}_i + F_N. \tag{9}$$

### *C. Loss Function*

The relative ranking relation between the samples within each batch is helpful for further optimization of the model. Therefore, we use the combined loss, the overall learning objective is defined as

$$\mathcal{L}_{\text{total}} = L_{\text{quality}} + \alpha L_{\text{relative}} \tag{10}$$

where $L_{quality}$ and $L_{relative}$ are the loss terms for the MSE loss [36] and the relative ranking loss, respectively. And $\alpha$ is a weight coefficient. To benefit from the features extracted by the convolutional network, we use a fully connected (FC) layer as a fusion layer to map the above features and predict the perceived quality of the images. We minimize the MSE loss to train our network.

$$\mathcal{L}_{\text{quality}} = \frac{1}{N} \sum_{i=1}^{n} (s_i - \hat{s}_i)^2 \tag{11}$$

where $s_i$ and $\hat{s}_i$ denote the ground-truth subjective quality score and predicted quality scores for the $i$-th image, respectively. While Mean Squared Error (MSE) loss is a widely used and effective approach for image quality prediction tasks, it falls short in capturing the relative ranking and correlation

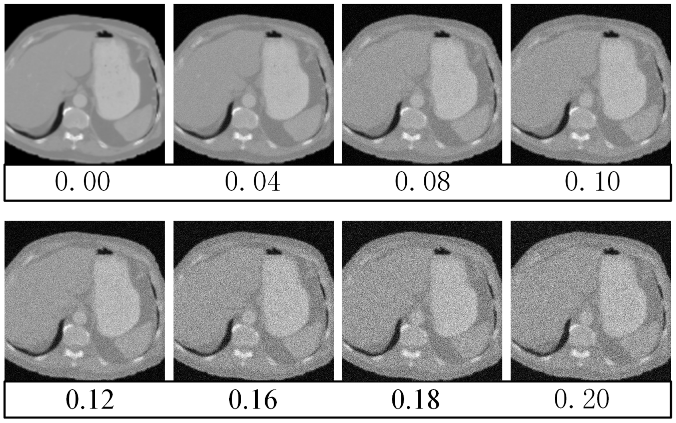


Fig. 3. Visualization of several example images with different score labels in the abdominal CT dataset. The figure shows CT images with different noise levels and the labels corresponding to the images from top to bottom.

between images, which is essential for accurate quality assessment. Here, our goal is to consider the relative ranking relationships between each batch of samples. However, considering the relative ranking information of all samples is computationally expensive. To balance performance and efficiency, we apply this relative ranking mechanism selectively, focusing only on extreme cases where the ranking discrepancy is most critical. The relative ranking loss is defined as:

$$\mathcal{L}_{\text{relative}} = \mathcal{L}_{\text{triplet}}\left(q_{\max}, q'_{\max}, q_{\min}\right) + \mathcal{L}_{\text{triplet}}\left(q_{\min}, q'_{\min}, q_{\max}\right) \tag{12}$$

$$= \max\left\{0, d\left(q_{\max}, q'_{\max}\right) - d\left(q_{\max}, q_{\min}\right) + \beta_1\right\} \tag{13}$$

$$+ \max\left\{0, d\left(q'_{\min}, q_{\min}\right) - d\left(q_{\max}, q_{\min}\right) + \beta_2\right\}. \tag{14}$$

In the images within each batch, let $q_{\max}$ ,$q'_{\max}$ ,$q_{\min}$ ,$q'_{\min}$ denote the predicted quality for images with highest, second highest, lowest, and second lowest subjective quality scores. $s_{q_{\max}}$ denotes the subjective quality score corresponding to the image with the predicted quality score $q_{\max}$ , and a similar notation rule applies to the rest. We set $\beta_1 = s_{q'_{\max}} - s_{q_{\min}}$ and $\beta_2 = s_{q_{\max}} - s_{q'_{\min}}$ .

## IV. Experiments

### A. Implementation Details

To train the PFGM model, we train the network $100k$ iterations with $D = 64$, a learning rate of $2 \times 10^{-4}$, and a batch size of 32. Upon completing the training process, we employ the trained PFGM to obtain the primary contents from the LDCTIQA2023 dataset and abdominal CT dataset. For the hierarchical feature fusion regression network, we select ResNet50 as the backbone. The network is trained with a learning rate of $4 \times 10^{-5}$ and a batch size of 16. A StepLR scheduler is employed with a step size of 25 epochs and a decay factor of 0.7, alongside Exponential Moving Average (EMA) regularization with a decay rate of 0.997. All experiments are carried out on an NVIDIA GeForce RTX 3090 utilizing PyTorch for both training and testing.

### B. Datasets

**Mayo Low-dose CT Datasets [37]:** This publicly available clinical dataset contains CT images from 10 patients and is used exclusively for training the Poisson Flow Generative Models (PFGM), which does not contain human perception scores for each image. The data is strategically split into a training set comprising the first 8 patients, totaling 4800 slices, and a validation set containing the last 2 patients, with 1136 slices.

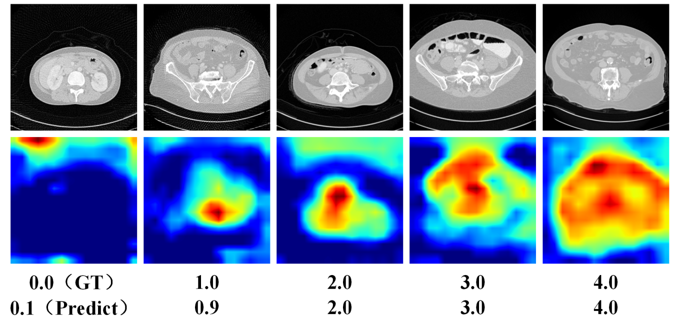


Fig. 4. Visualization of several example images from the LDCTIQA2023 test dataset. From top to bottom are the distorted image, heatmap, GT, and the scores predicted by our model.

TABLE I
Comparisons on the LDCTIQA2023 Dataset. The best and the second best values are marked with **Bold** and Underline.

| Methods | PLCC | SROCC | KROCC | Overall |
|---|---|---|---|---|
| DBCNN [39] | 0.9716 | 0.9693 | 0.8692 | 2.8101 |
| HyperIQA [40] | 0.9680 | 0.9694 | 0.8672 | 2.8046 |
| QPT [41] | 0.9743 | 0.9732 | 0.8797 | 2.8272 |
| TReS [42] | 0.9755 | 0.9745 | 0.8786 | 2.8286 |
| AHIQ [43] | 0.9762 | 0.9746 | 0.8810 | 2.8318 |
| MANIQA [8] | 0.9768 | 0.9786 | 0.8891 | 2.8445 |
| MD-IQA [26] | 0.9789 | 0.9792 | 0.9041 | 2.8622 |
| D-BIQA [13] | 0.9814 | 0.9816 | **0.9122** | 2.8752 |
| Ours | **0.9815** | **0.9837** | 0.9113 | **2.8765** |

**LDCTIQA2023 [38]:** LDCTIQA2023 includes a total of 1000 distorted abdominal images that exhibit both noise and streak artifacts. These images are generated at four different dose levels: 100%, 50%, 25%, and 10%. Of the total images, 900 are assigned to the training phase, while the remaining 100 are reserved for the testing phase. The final human perception score for each image is determined by averaging the individual ratings of five skilled radiologists.

**Abdominal CT Dataset:** The scarcity of datasets for CT image quality assessment has significantly hindered research and the generalization of experiments in this field. To overcome this limitation, we propose the creation of a novel CT image dataset specifically designed for IQA. Inspired by the PFGM training process, where different levels of noise are added to images ( $\hat{y}_i = y_i + R_i v_i$, where $y_i$ is the input image and $R_i v_i$ represents additive noise ), followed by denoising, we adopt a similar method. Specifically, we used different sigma values to control the level of Gaussian noise added to 512 original full-dose abdominal CT images. Under the guidance of professional radiologists, we selected 10 sigma values, ranging from 0 to 0.2 with an interval of 0.02. The labels of the dataset correspond to these sigma values, plus a label of 0 for the full-dose CT images, resulting in a total of 11 labels. Of the 5632 images, 5120 are used for training, while the remaining 512 are reserved for testing. Figure 3 shows some examples of images with different labels.

### C. Evaluation Criteria

Following the prior works, we use Pearson's linear correlation coefficient (PLCC), Spearman's rankorder correlation coefficient (SROCC), and Kendall rank-order correlation coefficient (KROCC) as the metrics to evaluate the performance of our models. The PLCC is defined as follows

$$\text{PLCC} = \frac{\sum_{i=1}^{N}(s_i - \mu_{s_i})(\hat{s}_i - \mu_{\hat{s}_i})}{\sqrt{\sum_{i=1}^{N}(s_i - \mu_{s_i})^2}\sqrt{\sum_{i=1}^{N}(\hat{s}_i - \mu_{\hat{s}_i})^2}}, \tag{15}$$

TABLE II

COMPARISONS ON THE ABDOMINAL CT DATASET.THE BEST AND THE SECOND BEST VALUES ARE MARKED WITH **BOLD** AND UNDERLINE.

| Methods | PLCC | SROCC | KROCC | Overall |
|---|---|---|---|---|
| TReS | 0.7150 | 0.7149 | 0.5004 | 1.9303 |
| MANIQA | <u>0.9976</u> | 0.9467 | 0.8526 | 2.7969 |
| AHIQ | **0.9981** | **0.9957** | 0.9529 | **2.9467** |
| Ours | 0.9950 | <u>0.9559</u> | **0.9658** | <u>2.8167</u> |

where $s_i$ and $\hat{s}_i$ respectively indicate the ground-truth and predicted quality scores of the $i$-th image, $\mu_{s_i}$ and $\mu_{\hat{s}_i}$ indicate the mean of them. $N$ denotes the number of testing images. The SROCC is defined as:

$$\mathrm{SROCC} = 1 - \frac{6\sum_{i=1}^{N} d_i^2}{N(N^2-1)} \tag{16}$$

where $d_i$ denote the difference between the ranks of the $i$-th test image in ground-truth and predicted quality scores. The KROCC is defined as:

$$\mathrm{KROCC} = \frac{2(C-D)}{N(N-1)} \tag{17}$$

where $C$ is the number of consistent pairs, indicating that the relative order of the two variables is consistent when compared pairwise. $D$ is the number of inconsistent pairs, indicating that the relative order of the two variables is inconsistent when compared pairwise.

The overall metric is defined as:

$$\mathrm{Overall} = PLCC + SROCC + KROCC \tag{18}$$

### D. Results

We conduct both quantitative and qualitative comparisons of our method against several state-of-the-art methods, including CNN-based approaches (DBCNN [39], HyperIQA [40] and QPT [41]), Transformers based approaches (TReS [42], AHIQ [43], MANIQA [8], MD-IQA [26]) and denoising diffusion probabilistic model based approach D-BIQA [13].

In TableI, we provide a detailed comparison of our method with these state-of-the-art approaches on PLCC, SROCC, KROCC, and Overall. From Table I, we can find that the proposed method outperforms the state-of-the-art methods on LDCTIQA2023 databases for SROCC, PLCC and Overall evaluation metrics. Compared to the three CNN-based methods, our method achieves a significant performance improvement, boosting overall scores by at least $5\%$. Furthermore, even against more advanced Transformer-based methods, our model enhances total scores by $1\%$ to $5\%$. Notably, in comparison to D-BIQA, which also utilizes generative models to obtain corresponding high-quality images, our approach also surpasses D-BIQA across all three evaluation metrics. And the image generation speed of PFGM in our method is nearly $490$ times shorter than that of the general diffusion model, while the model's training parameter count is also significantly reduced, from $271.01M$ to $92.66M$, a reduction of approximately 2.9 times, as shown in Table III.

To further demonstrate the effectiveness of our method, we also conduct experiments on the Abdominal CT datasetas shown in Table II. Compared to other state-of-the-art methods, our method achieves the best performance in KROCC and secures the second-best results in SROCC and Overall metrics. While AHIQ demonstrates strong overall performance, it heavily relies on reference images during the training phase. In contrast, our method achieves comparable or superior results without requiring reference images, showcasing its greater robustness and adaptability.

In Figure 4, we present the distorted images and corresponding heatmaps for five representative examples from the low-dose CT BIQA test dataset. These heatmaps clearly highlight the salient regions, which are crucial areas that heavily influence human visual perception. When viewing medical images, human attention naturally gravitates toward certain features, and these areas play a dominant role in shaping overall perception. Our model effectively captures these high-impact regions, reflecting its sensitivity to the critical aspects of image quality.

TABLE III

ABLATION STUDY OF GENERATIVE MODELS, WHERE AIT REPRESENTS THE AVERAGE INFERENCE TIME FOR GENERATING A SINGLE IMAGE.

| Generative Models | Overall | Time(s/image) | Params |
|---|---|---|---|
| diffusion model [44] | 2.7767 | 94.531 | 271.01M |
| PFGM [33] | 2.8765 | 0.192 | 92.66M |

TABLE IV

ABLATION STUDY OF THE INDIVIDUAL COMPONENTS.

| PFGM | HFFRN | Lr | Metrics | | |
|---|---|---|---|---|---|
| | | | SROCC | KROCC | Overall |
| | | | 0.9752 | 0.8914 | 2.8439 |
| ✓ | | | 0.9794 | 0.9051 | 2.8646 |
| | ✓ | | 0.9816 | 0.9043 | 2.8680 |
| ✓ | ✓ | | 0.9816 | 0.9089 | 2.8719 |
| ✓ | ✓ | ✓ | **0.9837** | **0.9113** | **2.8765** |

### E. Ablation Study

To investigate the effectiveness of each component of the proposed method, we conduct the ablation study experiments on the dataset LDCTIQA2023 [38], as shown in Table IV. We first add posterior sampling Poisson flow generative models (PFGM) and process the pseudo-reference image generated by PFGM as the network input. This led to notable performance gains, with SROCC and KROCC increasing by 0.6% and 1.3%, respectively.

Then, we use a pretrained ResNet50 as our backbone model and analyze the effect of each individual component by comparing SROCC, KROCC, and Overall. The results are shown in Table IV. We first examine the effectiveness of our proposed hierarchical feature fusion regression network (HFFRN) by concatenating them with ResNet50 output features. The SROCC and KROCC obviously improved with around $0.4\%$ and $1.4\%$. When both PFGM and HFFRN are added, the performance is further improved. At last, we add relative ranking loss (Lr) on this basis, SROCC and KROCC further improved to the highest value of $98.37\%$, $91.13\%$, and Overall reaches $2.8765$.

## V. CONCLUSION

In this work, our proposed PFGM-IQA introduces a novel approach to enhancing image quality assessment (IQA) for CT scans by leveraging Poisson Flow Generative Models and a hierarchical feature fusion regression network. PFGM is shown to be highly effective in creating pseudo-reference images that significantly improve the performance of regression models in capturing intricate image features. Furthermore, the integration of ViT and CNN within our hierarchical fusion network enables the seamless combination of non-local and local features, leading to highly accurate quality predictions. The synthesis of a comprehensive dataset tailored for CT quality assessment further highlights the potential of our approach to contribute to future research in this domain. Our results demonstrate the efficacy of the proposed framework, offering a new pathway for advancements in both medical image quality assessment and related clinical applications.

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